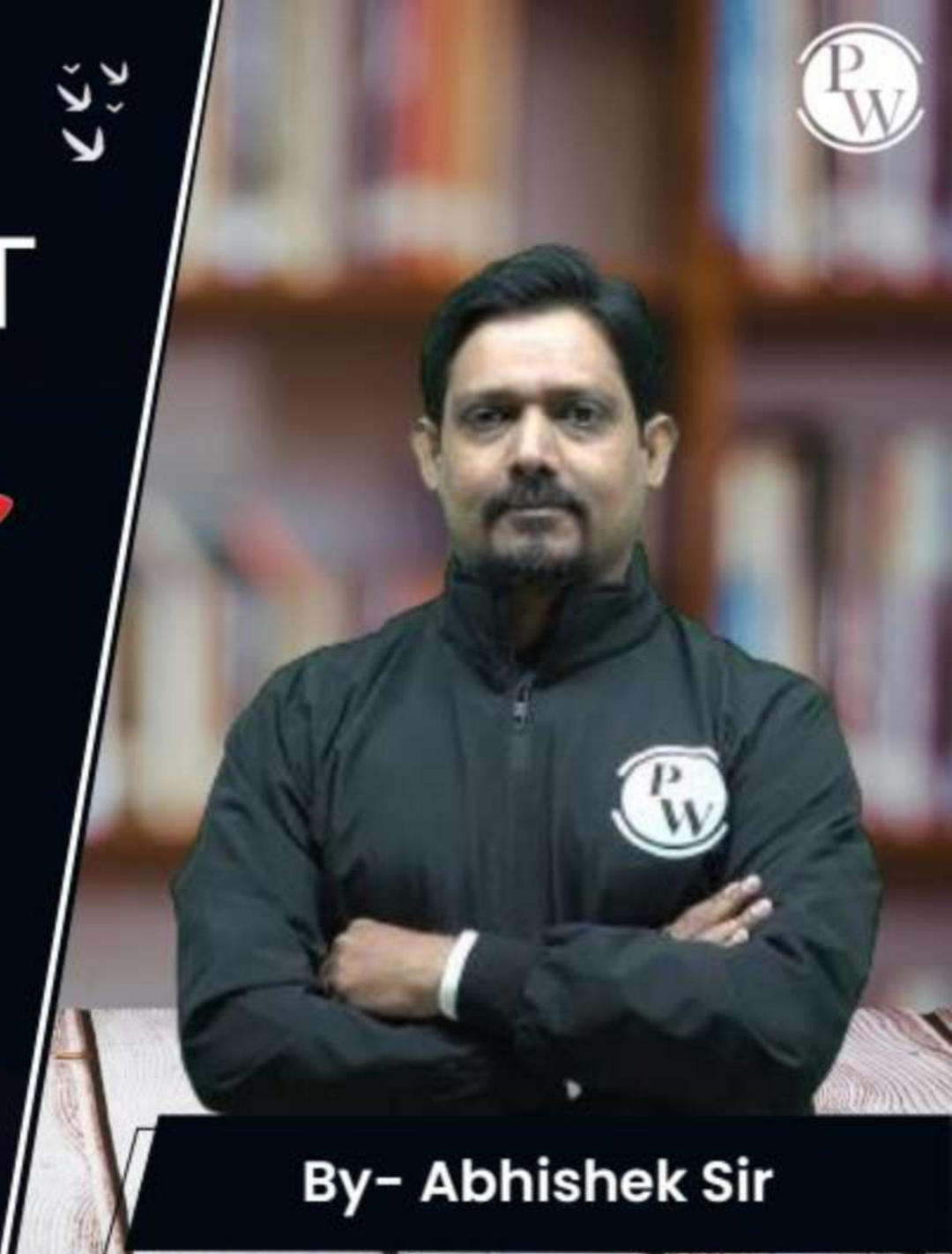
Computer Science & IT



C programming

Array & Pointers

Lecture No. 03



By- Abhishek Sir

Recap of Previous Lecture



Topic — Array

Topic — pointer Arithmetic

Topic — Addition & Subtraction with constant

Topic — ++, --

Topic

Slide

Topics to be Covered



- Topic — How to access array element
- Topic — Array data type (Array Niwas)
- Topic — A[i] expansion
- Topic — Array declaration
- Topic

Slide



Array

Collection of element

* Similar Data type
* Arranged in Sequential manner



Array

A[0]	A[1]	A[2]	A[3]
11	12	13	14
100	104	108	112

int A[4] = {11, 12, 13, 14};

Array starting Address

100
101
102
103

Base Address : 100

* Size of each element : 4B

printf ("%d", A[0]);

A[1]
A[2]
A[3]

Slide



Array

A[0] → Index starts from 0

int a[100]

startIndex 0

endIndex 99

float value will result in Error

Slide



Array

How Indexing works

Expansion of A[i] ⟶ *(A+i)

Name of array represent Address of first element of Array



Array

int A[4] = {11, 12, 13, 14}

A[0]	A[1]	A[2]	A[3]
11	12	13	14
100	104	108	112

Array of integer

A Address of first element ≡ A address of integer

Slide



Array

A[0]	A[1]	A[2]	A[3]
11	12	13	14
100	104	108	112

$A[0] = *(A+0)$

$= *(100+0)$

$= *(100) = 11$

$A[1] = *(100+1)$

$= *(100+1*4)$

$= *(100+4)$

$= *(104) = 12$

$A[3] = *(100+3)$

$= *(100+3\times4)$

$= *(112) = 14$

Slide



Array

$$ptr + c = ptr + c * \text{Size of data type}$$

$$ptr - c = ptr - c * \text{Size of data type}$$



Question

```
include<stdio.h>
int main()
{
int a[] = {1,23,4,5};
int *b= a+3;

printf("%d " b[-2]);
}
```

(A) 1
(B) 23
(C) 4
(D) 5

// b[-2] = 100. update

Slide



Question

```
include<stdio.h>
int main()
{
int a[] = {1,23,4,5};
int *b= a+3;

printf("%d " b[-2]);
}
```

1	23	4	5
100	104	108	112

$a+3 = 100+3 = 100+3\times4 = 112$

b [112]

$b[-2] = *(b-2)$

$= *(112-2)$

$= *(112-2\times4) = *(104)$

$= 23$

Slide

`int a[];` Ammount of memory allocate Not known — Not allowed

`int a[] = {1,2,3,4};` // Size of array is 4 int

Hence : 16 Bytes

Array Size = 16 Byte

`int a[4] = {1,2};` `printf("%d", a[3]);` //output 0

Rest of element automatically assigned to 0

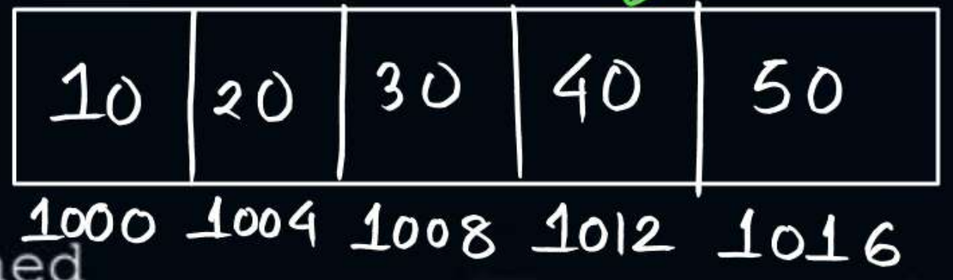
There is No check/error for violation of Array Bound

Question

Assume that array elements are stored from 1000, 1004, 1008, 1012, 1016

```
int main (){
      int a[] ={10,20,30,40,50};
      int i,*b ;
      b = a+4;          //1000 is assigned
      printf("%d", b[-1]);
      return 0;
}
```

The value printed by the program is_____?



b = a+4 = 1000+4 = 1000+4×4
= 1016

b[-1] = *(1016-1)
= *(1016-1×4)
*(1012) = 40

Slide



Difference Between two Address

$$ptr_2 = ptr_1 + c$$

$$ptr_2 = ptr_1 + c * \text{Size of data type}$$

$$\Rightarrow ptr_2 - ptr_1 = c * \text{Size of data type}$$

$$\Rightarrow \boxed{c = \frac{ptr_2 - ptr_1}{\text{Size of data type}}}$$

Slide



Difference Between two Address

* Imp

$$\underline{ptr_2 - ptr_1} = \frac{ptr_2 - ptr_1}{\underline{\text{Size}} \text{ of data type}} = \text{Constant}$$

Same pointer type

2024 ←

Imp

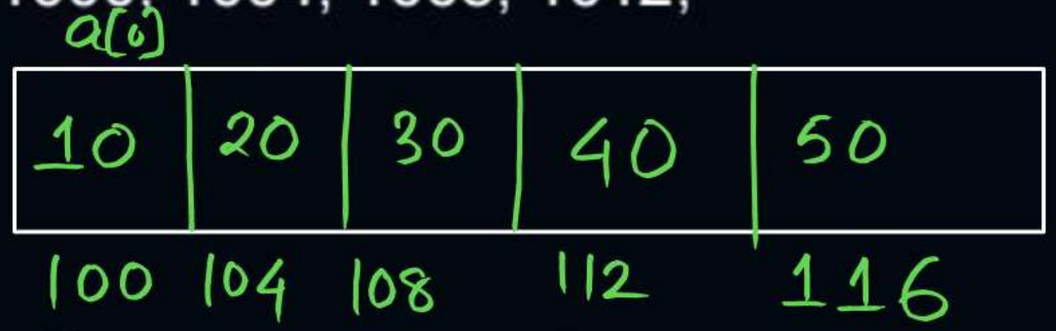
Slide

Question

Assume that array elements are stored from 1000, 1004, 1008, 1012, 1016

```
int main (){
        int a[] ={10,20,30,40,50};
        int i,*b,*b1;
        b = a;          //1000 is assigned
        b1 = a+4        //1000 +4 ×4 = 1016 assigned
        i = b1-b;
        printf("%d",i);
        return 0;
}
```

The value printed by the program is 4 ?



b = a ; = b 100

$b_1 = a+4 = 100+4 = 100+4\times4$

$= 116$

$i = 116-100$

$= \frac{116-100}{4} = \frac{16}{4} = 4$

Slide



Question

Assume that array elements are stored from 1000, 1004, 1008, 1012, 1016

```
int main (){
      int a[] ={10,20,30,40,50};
      int i,*b,;
      b = a+4;
      printf("%d", b[-1]);
      return 0;
}
```

The value printed by the program is_____?



Array

```
#include <stdio.h>
int main(){
    int a[] = {1,2,3,4};

    printf("%u\n", a);
    printf("%d\n", *a);
    printf("%u\n", &a);

    printf("%u\n", a+1);
    printf("%d\n", *a+1);
    printf("%u\n", &a+1);

    return 0;
}
```

A[0]	A[1]	A[2]	A[3]
11	12	13	14
100	104	108	112



Arithmetic Operation with Pointer

```
#Q
main (){
      int a[] ={10,20,30,40,50};
      int i,*b;
      b = &a[4]-4; // 116-4
      for(i=0;i<=4;i++){
            printf("%d",*b);
            b++;
      }
return 0;
}
Output_____
```

a	a[1]	a[2]	a[3]	a[4]
10	20	30	40	50
100	104	108	112	116

a is address of first element

& a[4] = 116

116 - 4*4 = 116 - 16 = 100

b = 100

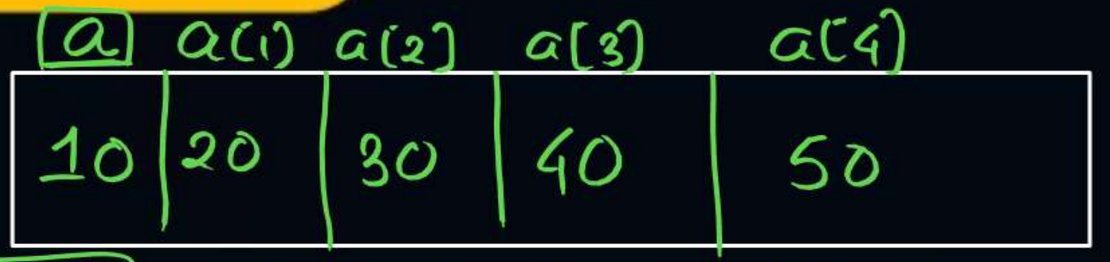
Slide

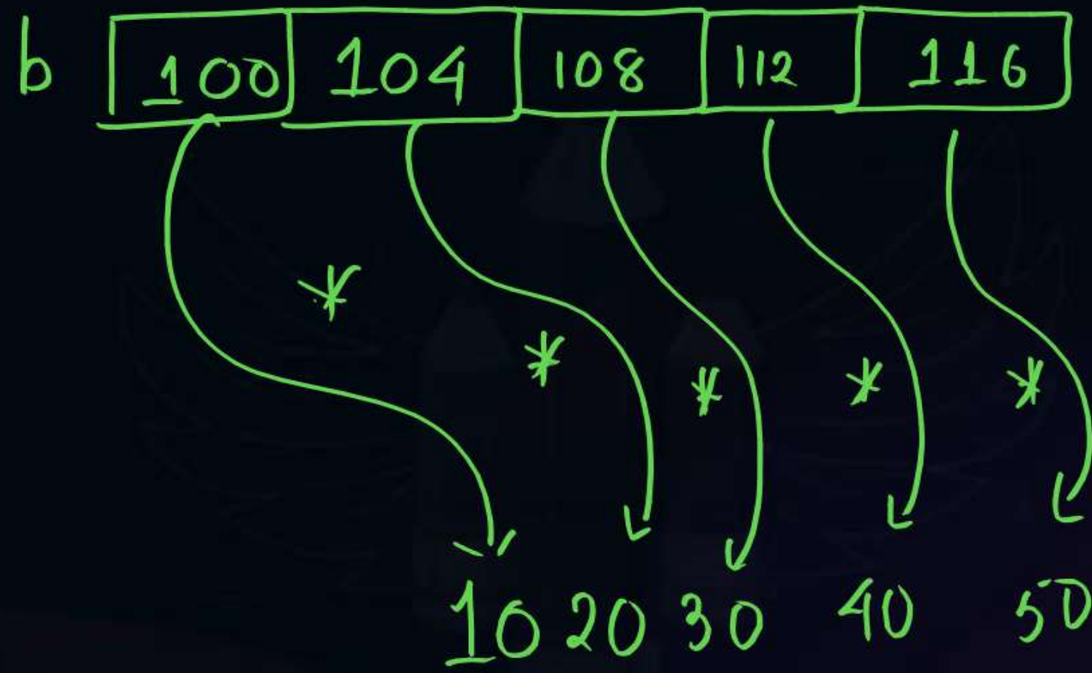
Arithmetic Operation with Pointer

#Q

```
main (){
      int a[] ={10,20,30,40,50};
      int i,*b;
      b = &a[4]-4;
      for(i=0;i<=4;i++){
            printf("%d",*b);
            b++;
      }
return 0;
}
```

Output 1020304050



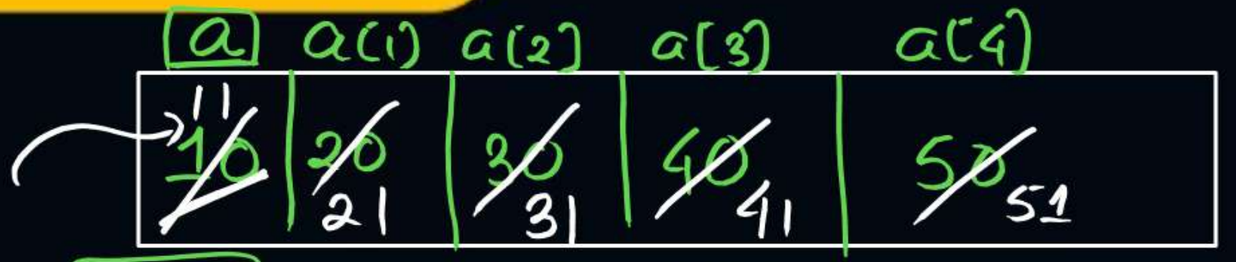


Slide

Arithmetic Operation with Pointer



```
#Q
main (){
      int a[] ={10,20,30,40,50};
      int i,*b;
      b = &a[4]-4;
      for(i=0;i<=4;i++){
            printf("%d", ++*b);
            b++;
      }
return 0;
}
Output_____
```



++*b

*(100) *(104)

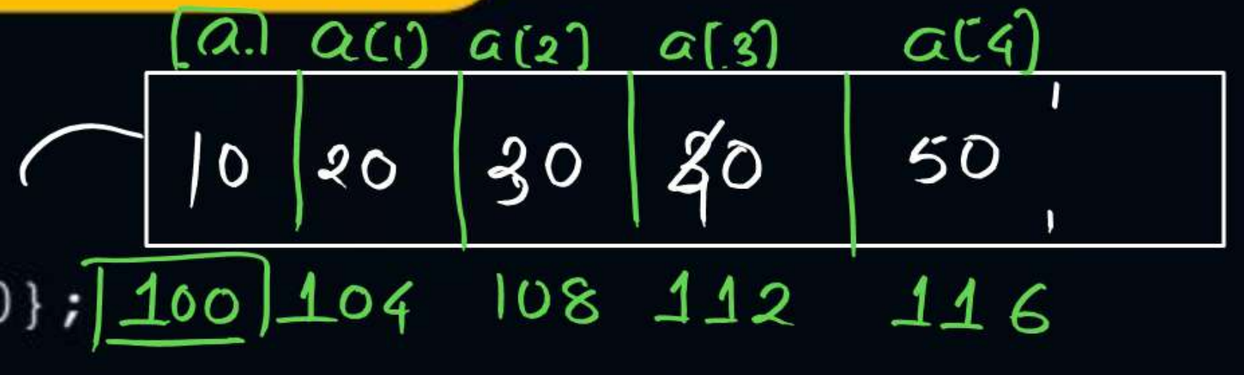
( )

output

1|2|3|4|51

Slide

Arithmetic Operation with Pointer




```
#Q
main (){
      int a[] ={10,20,30,40,50};
      int i,*b;
      b = &a[4]-4;
      for(i=0;i<=4;i++){
            printf("%d", ++*b);
            b++;
      }
return 0;
}
Output_____
```

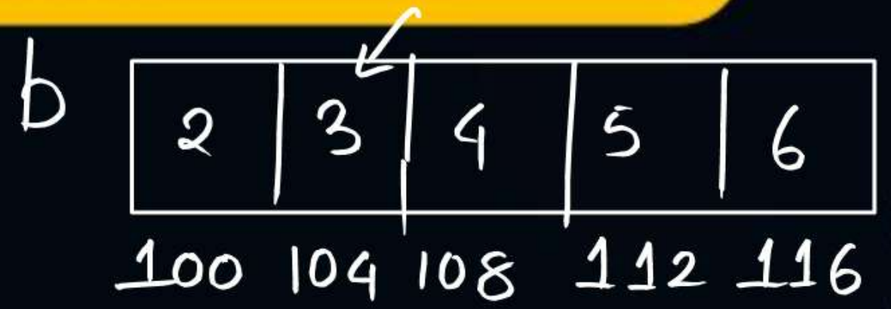
(*b)++;

Slide

Question

Out put of the program

```
#include<stdio.h>
     int main(){
    int *i , b[] = {2, 3, 4, 5, 6};
     i =b;
     *++i;
     printf ("%d\t" ,*i);
     i=i+3;
      printf ("%d\t" ,*i);
     }
```

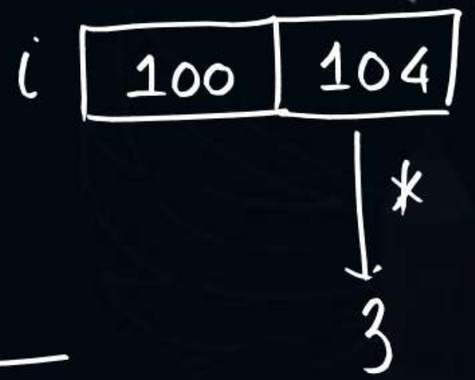


3 6

(A) 2, 3
(B) 3, 4
(C) 3, 6
(D) Error

tab

i = b;



i = i+3 = 104+3 = 104+3×4
= 104+12 = 116

Slide

Question

#Q What is the output of the following program ?

```
#include<stdio.h>
int main(){
int i , b[] = {2, 3, 4, 5, 6}, *p ;
p = b ;
b++;
printf ("%d\t" ,*b) ;
}
```



(A) 1
(B) 2
(C) 3
(D) Error (

b 100

p = b =

p 100

Constant ++ Not allowed L-value

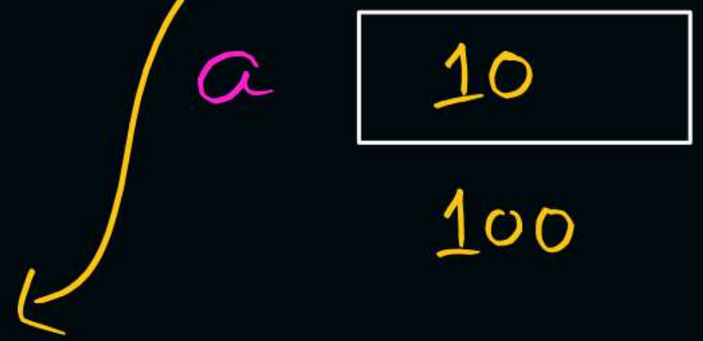
The association of Name of array & its starting Address is constant Associa~

Slide

r-value : value present in that Address

L-value is address of variable

```
int a,b;
a=10;
b=a;
```



=a;

↑

Right side of assignment



location of a updated by 10

variable used in Left side of assignment

a =

↑ Address of a

5++; Error

↑

5 is constant

↑

constant value has No Location



Question

PYQ

2019

PYQ

Consider the following C program:

```
#include <stdio.h>
int main(){
      int arr[]={1,2,3,4,5,6,7,8,9,0,1,2,5}, *ip=arr+4;
      printf("%d \n", ip[1]);
      return 0;
}
```

The number that will be displayed on execution of the program is _____.

(A) 4
(B) 5
(C) 6
(D) 7

*(ip+1)

*(116+1) : *(120) = 6

100+4 = 100+4×4 = 116

100	104	108	112	116	120	124	128	132	136	140	144	148
1	2	3	4	5	6	7	8	9	0	1	2	5



Question

#Q What is the output of the following program ?

```
#include<stdio.h>
int main(){
int i , b [] = {2, 3, 4, 5, 6}, *p ;
p = b ;
++*p ;
printf ("%d\t" ,*p) ;
p += 2 ;
printf ("%d" , *p);
}
```

(A) 1 3
(B) 2 3
(C) 2 4
(D) 3, 4

Slide



Question

Consider the following C program

```
#include <stdio.h>
int main(){
   int a[] = {2, 4, 6, 8, 10};
   int i, sum = 0, *b = a + 4;
   for (i = 0; i < 5; i++ )
      sum = sum + (*b - i) - *(b - i);
   printf("%d\n",sum);
   return 0;
}
```

The output of the above C-program is _____.

Slide



Question

What is the output of the following C program?

```
#include <stdio.h>
int main() {
      double a[2]={20.0, 25.0}, *p, *q;
      p = a;
      q = p + 1;
      printf("%d,%d", (int)(q - p), (int)(*q - *p));
      return 0;
}
```

(A) 4,8
(B) 1,5
(C) 8,5
(D) 1,8



2 mins Summary

6:30 pm

Topic — pointer Arithmetic

Topic — A[i] = *(A+i)

Topic — Subtraction of pointer

Topic

Topic

Slide

THANK - YOU

